

1.Type of applied technical documents

Name	FANUC Robot series R-30iB/R-30iB Mate CONTROLLER iRVision 3D Lasser Vision Sensor Application OPERATOR'S MANUAL
Spec.No./Ed.	B-83304EN-2/02

2.Summary of Change

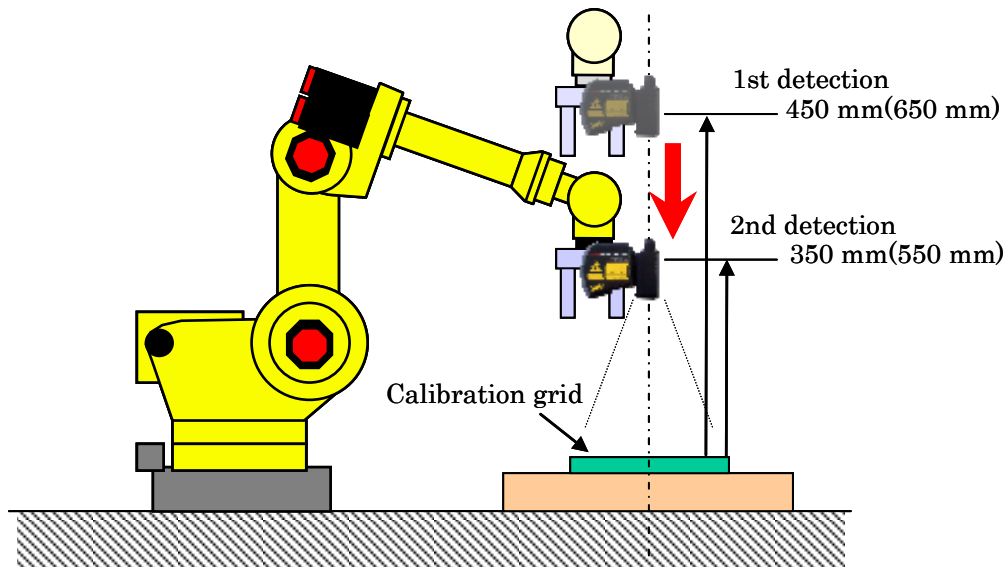
Group	Name/Outline	New, Add, Correct, Delete	Applicable Date
Basic			
Optional Function	Modify the following sections: 3.2.5 4.1.5, 4.2.4 5.1.5, 5.2.4 6.1.5, 6.2.4 7.1.5, 7.2.4	New	Immediately
Unit			
Maintenance Parts			
Notice			
Correction			
Another			

[illegible]

TEACHING EXAMPLE

3.2.5 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a robot-mounted camera, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



After one set of calibration tool is created with 3DL vision calibration, another set of calibration tool does not need to be created even after the camera Measurement Position is changed. This is because *iRVision* uses the current robot position when calculating the position of the workpiece.

Placing the 3D Laser Vision Sensor so that it faces the calibration grid

1. Place the 3D Laser Vision Sensor so that it faces the calibration plate.

After setting the user frame for the calibration grid, place the 3D Laser Vision Sensor at a distance of about 400mm from the grid, and set the orientation of the sensor visually so that the sensor and grid face each other.

[illegible]

Setting calibration parameters

3. Press F2[CREATE].

Run Stop Hold Abort

MAIN LINE 0 T2 ABORTED JOIN 10%

IRVision Setup Main - Camera Calibration Tools

Name	Comment
KCALIB1	
KCALIB2	

- 1. Camera Setup Tools
- 2. Camera Calibration Tools
- 3. Vision Process Tools
- 4. Application Setup Tools

[TYPE] CREATE EDIT [VTYPE] [ATTR] >

- After setting up the necessary frames and placing the 3D Laser Vision Sensor so that it faces the calibration grid, select [Camera Calibration Tools] from the setup main screen.

- Press F2[CREATE]. The following screen is then displayed:

5. Enter the calibration name.

Run Stop Load Save MAIN LINE 0 T2 ABORTED JOINT 10%

irVision Setup Main - Camera Calibration Tools

Create new vision data.

Type: 3DL Calibration Tool

Name:

Comment:

OK Cancel

6. Press F4[OK] to create.

- Select the calibration type. When calibration for the 3D Laser Vision Sensor, select [3DL Calibration Tool]. When calibration for a 2D fixed camera, select [Grid Pattern Calibration Tool] or [Simple 2-D Calibration Tool].
Select [3DL Calibration tool].

- Assign a unique name to the camera calibration tool.

[illegible]

Purge	Open	Hold	Cancel	MAIN LINE 0 T2 ABORTED JOIN						10%	
Run	1/0	Fwd	Rv								
IRVision Setup Main - Camera Calibration Tools											
Name								Comment			
X CALIB1											
X CALIB2											
[TYPE]	CREATE			EDIT		[VTYPE]		[ATTR]		>	

- [illegible]

Select [Light Output Signal Type]

To turn on and off the LED installed in front of the camera of the 3D Laser Vision Sensor in conjunction with 3D Laser Vision Sensor vision execution, specify the [Light Output Signal Type]. Depending on the material and status of the plane measured by the 3D Laser Vision Sensor, lighting required for detecting 2D features might adversely affect the detection of the laser slits.

This function can turn on the LED when, for example, 2D features are detected by one 3D measurement and the function can automatically turn off the LED when the laser slits are detected.

When the signal status is changed during vision execution, the original signal status is restored after the vision execution. LED control can therefore be performed during vision execution without adding commands to turn the LED on and off to the robot program.

From the following signal types, select a type suitable for the application:

- Not Used: The LED is not controlled in concert with vision detection.
- DO: DO is used for LED signal control.
- RO: RO is used for LED signal control.
- MUX: Select when LED is connected to a multiplexer.

When the DO or RO is selected as [LED Output Signal Type], set [Light Output Signal Number], [Light Output Signal Polarity], and [Light on Snap Delay]. In [Light Output Signal Number], specify the DO or RO number used for LED switching. [Light Output Signal Polarity] associates a signal state with the LED ON state. If [Signal=ON -> LED=ON] is set, for example, the LED is lit when DO or RO with a specified number is ON. In [Light on Snap Delay], enter a wait time required to change the LED ON state.

When the MUX is selected as [LED Output Signal Type], set [LED Channel] and [LED Intensity]. Select the channel number of LED connected to a multiplexer, and the brightness (1 - 16) of LED to turn on.

Whether the LED is to be on or off during 2D measurement and 3D measurement is set in relevant items separately.

11. Set [Exposure Time]

Adjust [Exposure Time] so that the grid pattern on the calibration grid is viewed clearly. If the image is too dark, set a longer exposure time (increase the value). If the image is too bright, set a shorter exposure time (decrease the value). Each time the exposure time is changed, a new image is captured.

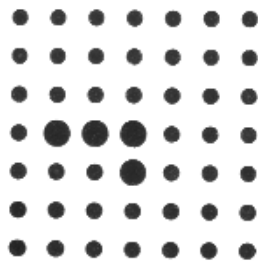
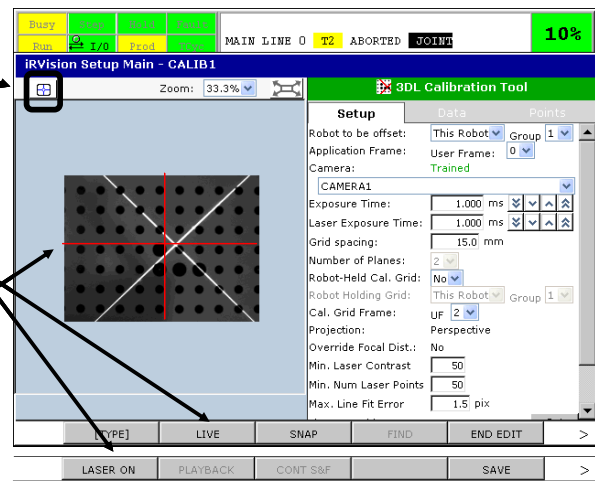


Fig. 3.2.5(a) Example of an image with appropriate exposure time

[illegible]

18. Move the robot so that the laser beam intersection and plate are displayed in the center of the image (crosshair).



- Jog the robot so that the grid is placed at the center of the image. The camera of the 3D Laser Vision Sensor and the calibration grid should be about 400mm apart, and the camera's line of sight should be perpendicular to the grid. Then, adjust the distance between the 3D Laser Vision Sensor and the grid so that the laser beam intersection is positioned at the center of the grid.

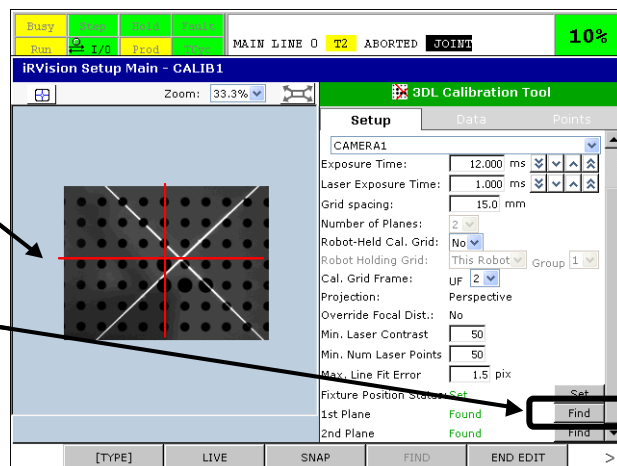
Parameters for detecting the laser points should be adjusted for calibration only when a correct detection cannot be made just by adjusting the exposure time. If the laser points are detected forcibly by changing the parameter values carelessly, the detection position may not be calculated correctly.

[illegible]

Detecting the 1st calibration plane

1. Teach the position in the robot program that places the laser beam intersection and plate in approximately the center of the image.

2. Execute the robot program, press F3[SNAP], and click [Find] for [1st plane].



1. Teach the position in the ...

Create a program as follows:

```

1: UFRAME_NUM=1 ;
2: J P[1] 100% FINE ;
3: PR[99]=LPOS
4: PR[99, 1]=0
5: PR[99, 2]=0
6: PR[99, 4]=0
7: PR[99, 5]=0
8: PR[99, 6]=0
9:
10: !Remove Backlash ;
11: PR[99, 3]=60
12: L P[1] 800mm/sec FINE ;
    : Offset,PR[99]
13:
14: !1st Plane ;
15: PR[99, 3]=50
16: L P[1] 800mm/sec FINE ;
    : Offset,PR[99]
17: PAUSE ;
18: ;
19: !2nd Plane ;
20: PR[99, 3]=(-50)
21: L P[1] 800mm/sec FINE ;
    : Offset,PR[99]
22: PAUSE ;
23: END

```

Specify the number set for [User Frame] in [Calibration Grid Frame].

Teach the robot position after movement.

Position taught for always snapping an image following movement in the same direction considering backlash.

Click the [Snap and Find] button for [1st Plane].

Click the [Snap and Find] button for [2nd Plane].

For the user frame number, specify the frame set for the calibration grid. After the robot has moved to an appropriate position for camera calibration, teach the position for P[1]. In the above sample robot program, [Set new id ?] appears on the teach pendant. Select F5[NO]. The calibration grid is actually measured at a position obtained by performing [Fixes Frame Offset] compensation for the taught position. Carefully operate the robot hand and camera not to cause them to interfere with the peripheral equipment and workpiece.

2. Execute the robot program, press F3[SNAP], and click [Find] for [1st plane]

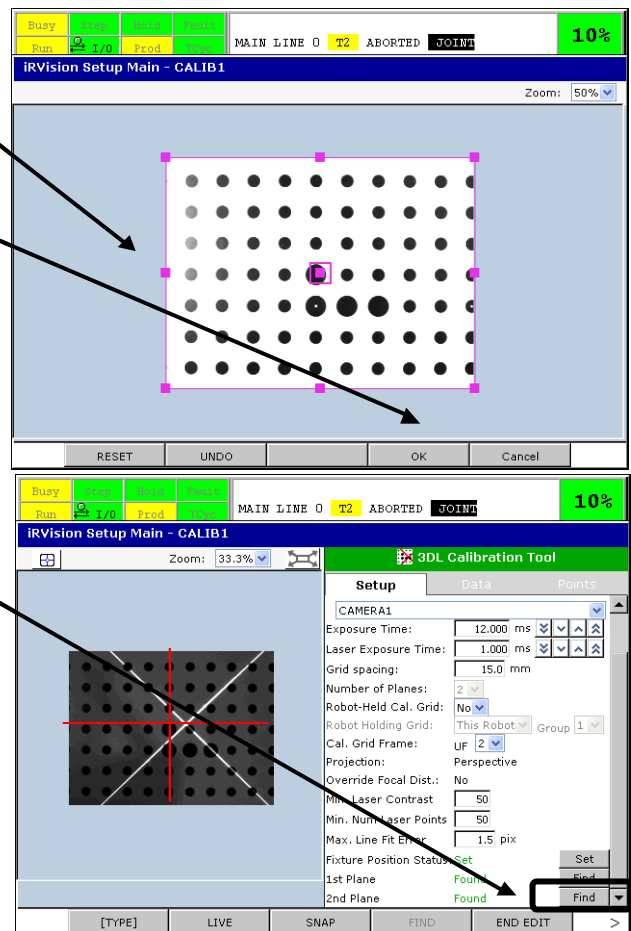
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Detecting the 2nd calibration plane

3. Teach the search window to enclose only perfect grids in the calibration grid.

4. Press F4 [OK].

5. Restart the program, press F3[SNAP], and click [Find] for 2nd Plane to teach the search window

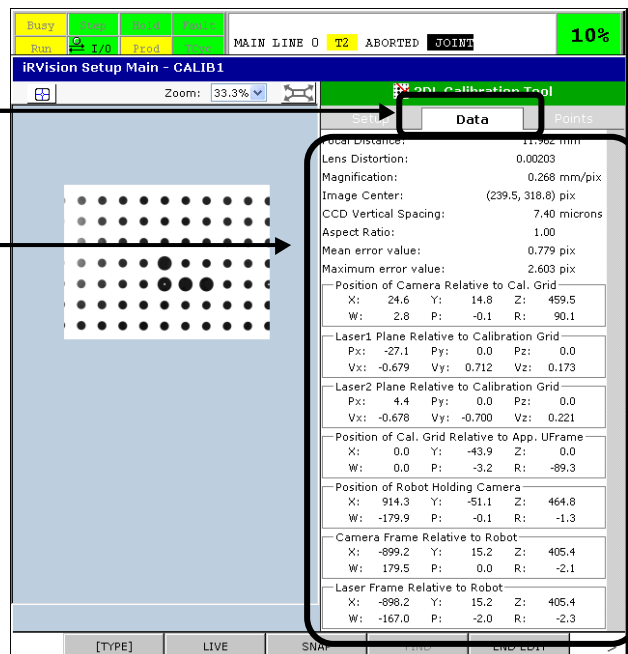
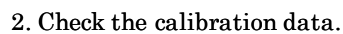


3. **Teach the search window to enclose only perfect grids in the calibration grid**
If the search range includes an area outside the calibration grid, a grid detection error could occur. So, teach the window so that only the grid pattern is enclosed.
4. **Press F4[OK]**
When modification of the window is completed, press F4[OK]. Check that all grids are detected and that the laser slits are detected clearly. If the detection is successful, [1st Plane] shows [Found].
5. **Restart the program, press F3[SNAP], and click [Find] for 2nd Plane to teach the search window**
Restart the robot program. After the program stops at the next Measurement Position, press F3[SNAP], and click the [Find] button for [2nd Plane] to make the second calibration plane detection. Check that all grids are detected and that the laser slits are detected clearly. If the detection is successful, [2nd Plane] shows [Found].

When [Status of fixture position] indicates that setting is done, and [1st Plane] and [2nd Plane] indicate that the calibration planes have been detected, 3DL calibration is completed. Check that [Status] for camera calibration shows [Trained].

[illegible]

Checking [Data]



When the [Data] tab is clicked, the screen display changes, and details on calibration can be checked.

If the calculated value in [Focal Distance] is the focal length of the lens used $\pm 5\%$, the value is considered to be permissible.

[Magnification] depends on the field of view at the time of calibration and the space between grids on the calibration grid. For calibration of a 3D Laser Vision Sensor equipped with an 8mm lens, between 0.35 and 0.45mm/pixel is normally obtained if calibration 1st plane is about 450mm away from the sensor and 2nd plane is about 350 mm away from the sensor.

[Image Center] indicates the image center calculated by calibration. The image center is located at around (240, 320). If the calculated center is within the range of this value $\pm 10\%$, it is permitted. If the indicated value is beyond this permissible range, confirm the calibration procedure, and perform calibration again.

[illegible]

Checking [Points]

1. Click [Points].

2. Check the detection point positions in the image.

3. Check the calibration point data.

4. Delete unnecessary points.

5. Save data and close the window.

1. **Click [Points]**
When the [Points] tab is clicked, the image display changes, showing details of detected calibration points and a slit detection image.
2. **Check the detection point positions in the image**

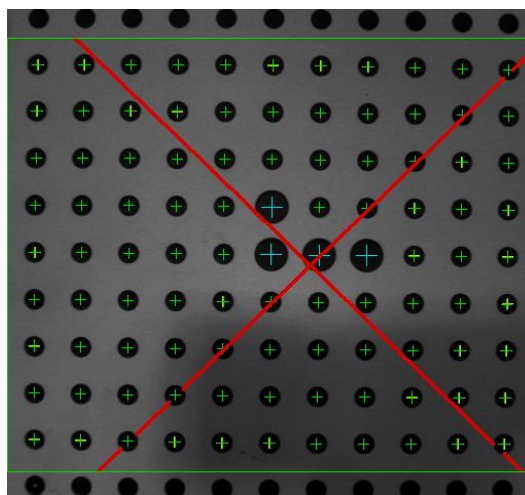


Fig. 3.2.5(e) Example of an image showing appropriate detection points

[illegible]

[illegible]

3. Select [Type].

4. Enter [Name].

5. Press F4[OK] for creation.

3. Select [Type]

Options vary depending on the installed software option. To create a vision process to check the precision of 3DL calibration, select [3DL Single-View Vision Process].

4. Enter [Name]

Assign a unique name to the vision process.

5. Press F4[OK] for creation

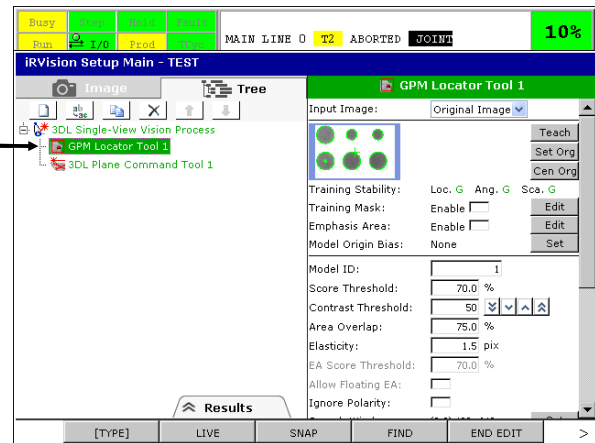
After selecting a type and entering a name, press F4[OK]. A precision check vision process is then created.

6. Select the created data, then press F3[EDIT].

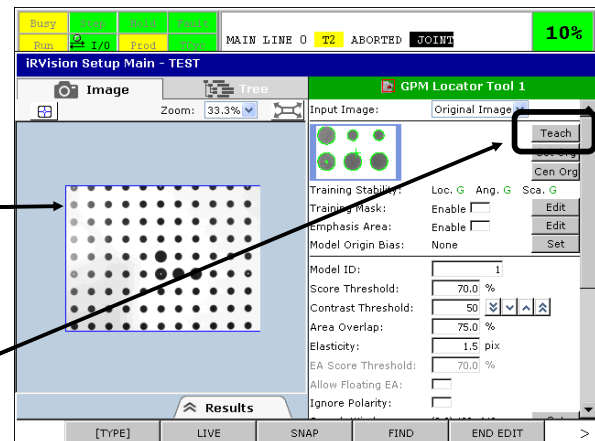
6. Select the created data, then press F3[EDIT]

Select Vision Process Tools for the created vision process, then press F3[EDIT]. The setup screen is then displayed.

								Title	Addition of the description about 3DL Calibration tool enhancement for FANUC Robot series R-30iB/R-30iB Mate CONTROLLER iRVision 3D Laser Vision Sensor Application OPERATOR'S MANUAL			
								Draw No.	B-83304EN-2/02			
Ed.	Date	Design	Description					FANUC CORPORATION		Page	15/40	
	Date	2018.3.30	Design	T.Satou	Apprv.	T.Andou						



From the tree window, select [GPM Locator Tool1].



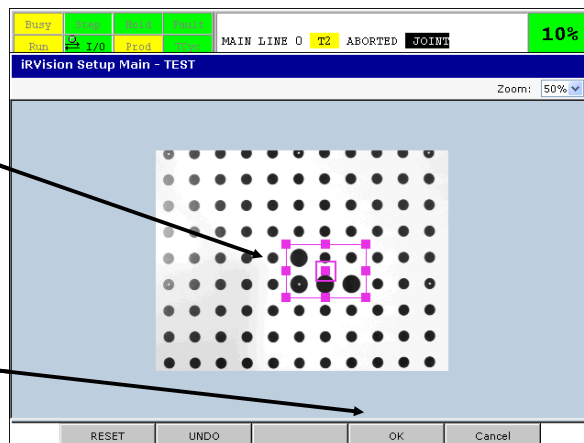
In the same way as for camera calibration, move the robot so that the laser intersection and the center of the calibration grid are placed in the center of the image. The robot is moved to the position taught in the program created for calibration.

When the [Teach Pattern] button is clicked, the GPM Locator Tool model teach screen is displayed.

[illegible]

14. Enclose large grids.

15. Press F4[OK].



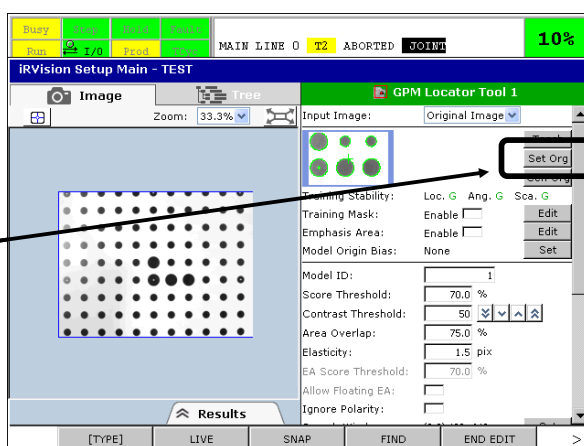
14. Enclose large grids

Change the area to enclose four large grids.

15. Press F4[OK]

When the F4[OK] is pressed, a model is taught, then the taught model is displayed.

16. Click [Set Origin].

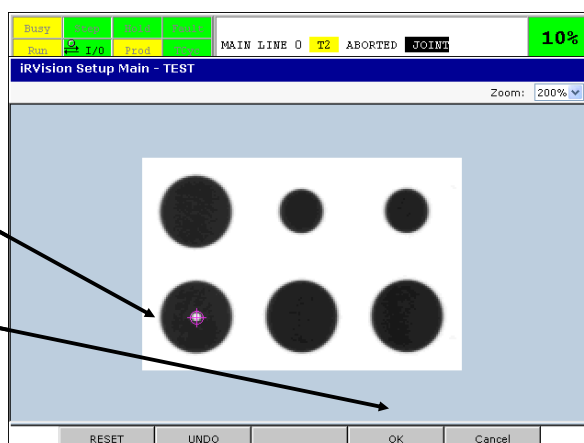


16. Click [Set Origin]

Change the origin of the model to match it with the origin of the frame set for the calibration grid.

17. Move the origin.

18. Press F4[OK].



17. Move the origin

						Title	Addition of the description about 3DL Calibration tool enhancement for FANUC Robot series R-30iB/R-30iB Mate CONTROLLER iRVision 3D Laser Vision Sensor Application OPERATOR'S MANUAL			
						Draw No.	B-83304EN-2/02			
Ed.	Date	Design	Description				FANUC CORPORATION		Page	18/40
	Date	2018.3.30	Design	T.Satou	Apprv.	T.Andou				

[illegible]

25. Press F3[SNAP], then F4[FIND]

Press F3[SNAP] , then F4[FIND]. An image is then captured, and a detection is made.

26. Take a note of measurement results

Take a note of the X, Y, Z, W, P, and R values in the measurement result obtained when the 3D Laser Vision Sensor and the calibration grid face each other, as reference data.

27. Move the robot position.

28. Press F3:[SNAP], then F4[FIND].

29. Compare the measurement results.

#	X	Y	Z	W	P	R	LaserID	Model ID	Score	Contrast	Fit Err.	Lean Angle
1	2.6	-44.6	0.0	-3.2	0.0	0.8	1	1	100.0	159.4	0.03	3.1

27. Move the robot position

Change the position and posture of the robot as far as the calibration grid is viewed within the field of view of the camera.

28. Press F3[SNAP], then F4[FIND]

Press F3[SNAP] , then F4[FIND]. An image is then captured, and a detection is made.

29. Compare the measurement results

Compare the measurement results with the reference data to confirm that they approximately match. When user frame 0 is specified in [Application Frame] in the set calibration tool, the output results should approximately correspond to the frame set for the calibration grid. When the frame set for the calibration grid is specified in [Application Frame] in the set calibration tool, all elements of the output results should be approximately 0.

30. Repeat the measurement

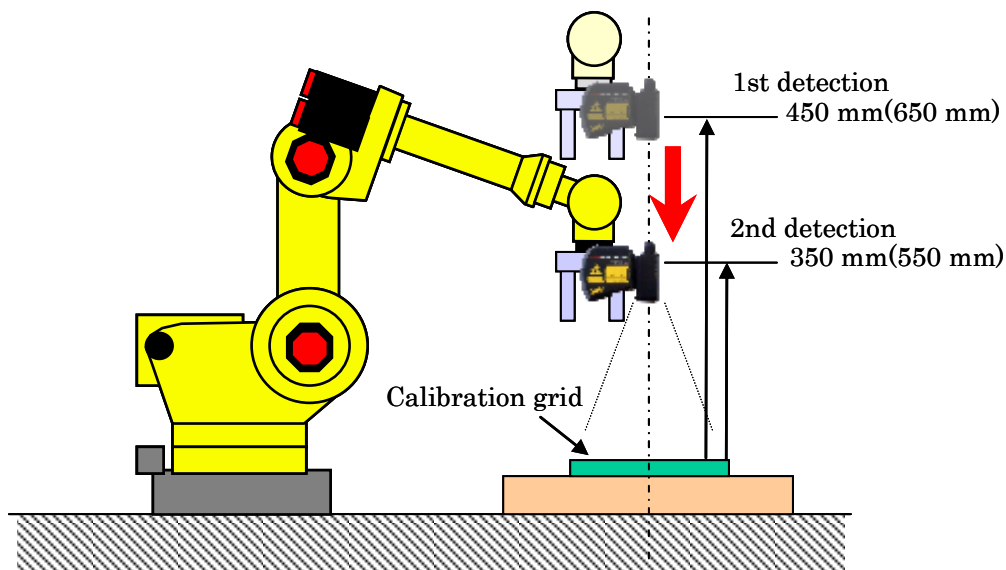
Repeat a sequence of operations from the movement of the robot to compare measurement results while changing the position and posture of the robot. If variations in results due to errors in the absolute position precision of the robot, precision of the 3D Laser Vision Sensor, and calibration precision are 0.5 mm or less for the X, Y, and Z elements or 1 degree or less for the W and P elements, calibration has been performed normally. If the results contain abnormal values, confirm the calibration procedure, and perform calibration again.

[illegible]

3DL SINGLE-VIEW VISION PROCESS

7.2.4 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a robot-mounted camera, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



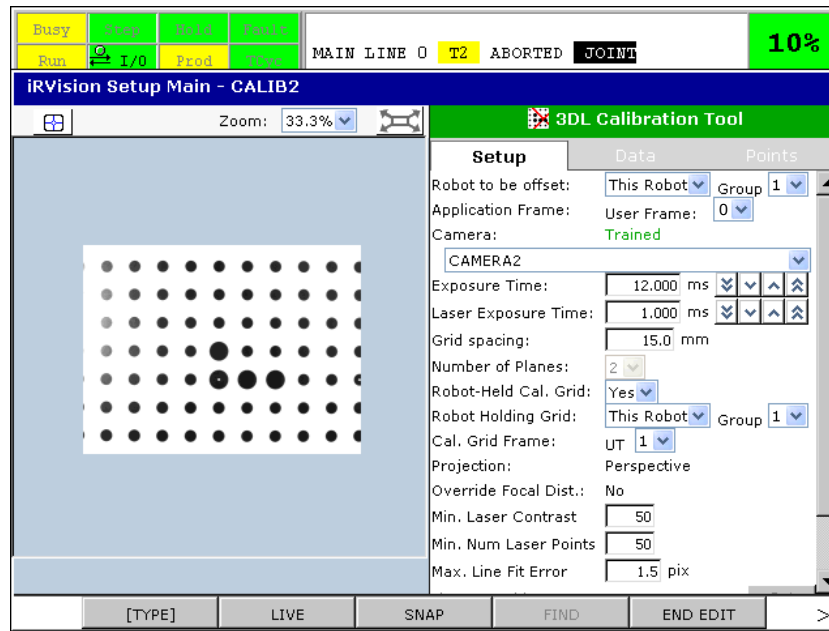
After one set of calibration tool is created with 3DL vision calibration, another set of calibration tool does not need to be created even after the camera Measurement Position is changed. This is because *iR*Vision uses the current robot position when calculating the position of the workpiece.

The teach screen for 3DL calibration is shown below. A calibration grid image is displayed.

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[illegible]

The teach screen for the 3DL calibration is shown below. A calibration grid image is displayed.

[illegible]

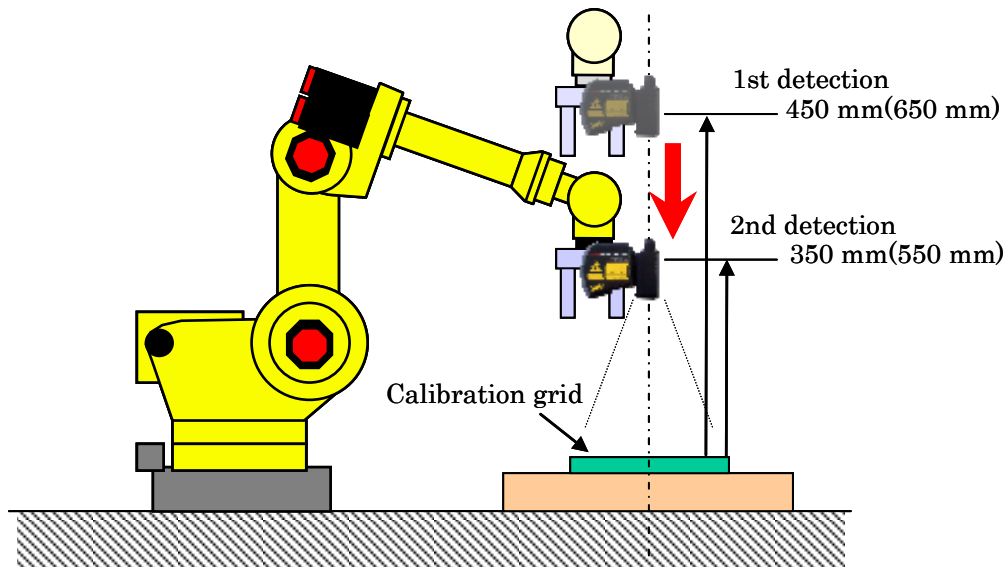
Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

[illegible]

3DL MULTI-VIEW VISION PROCESS

7.2.4 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a robot-mounted camera, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



After one set of calibration tool is created with 3DL vision calibration, another set of calibration tool does not need to be created even after the camera Measurement Position is changed. This is because *iRVision* considers the current robot position when calculating the position of the workpiece.

The teach screen for 3DL calibration is shown below. A calibration grid image is displayed.

[illegible]

The screenshot displays the **iRVision Setup Main - CALIB1** window. At the top, a status bar shows system information: **Busy**, **Stop**, **Reset**, **Pause**, **MAIN LINE 0**, **T2**, **ABORTED**, **JOINT**, and a **10%** progress indicator.

The main interface is divided into two primary sections:

- Left Panel:** A large area for visual feedback, currently showing a grid of black dots on a white background, representing the calibration target.
- Right Panel:** The **3DL Calibration Tool** configuration area, which includes a **Setup** tab and various adjustable parameters.

3DL Calibration Tool Setup Parameters:

- Robot to be offset:** This Robot (Group 1)
- Application Frame:** User Frame: 0
- Camera:** Trained
- CAMERA1** (Selected)
- Exposure Time:** 12.000 ms
- Laser Exposure Time:** 1.000 ms
- Grid spacing:** 15.0 mm
- Number of Planes:** 2
- Robot-Held Cal. Grid:** No
- Robot Holding Grid:** This Robot (Group 1)
- Cal. Grid Frame:** UF 2
- Projection:** Perspective
- Override Focal Dist.:** No
- Min. Laser Contrast:** 50
- Min. Num Laser Points:** 50
- Max. Line Fit Error:** 1.5 pix

At the bottom of the window, a navigation bar contains buttons for **[TYPE]**, **LIVE**, **SNAP**, **FIND**, **END EDIT**, and **EXIT**.

[illegible]

Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

[illegible]

3D Laser Vision Sensor

1st detection
350 mm(550 mm)

Calibration grid

2nd detection
450 mm(650 mm)

The screenshot displays the **iRVision Setup Main - CALIB2** window. At the top, a status bar shows system information: **Busy**, **Stop**, **Hold**, **Pause**, **Run**, **I/O**, **Prod**, **Stop**, **MAIN LINE 0**, **T2**, **ABORTED**, **JOINT**, and a green **10%** indicator.

The main window is divided into two panes. The left pane, titled **iRVision Setup Main - CALIB2**, contains a grid of black dots on a white background, used for calibration. The right pane, titled **3DL Calibration Tool**, contains a **Setup** tab with various configuration options.

The **Setup** tab includes the following settings:

- Robot to be offset:** This Robot (Group 1)
- Application Frame:** User Frame: 0
- Camera:** Trained
- CAMERA2** (selected)
- Exposure Time:** 12.000 ms
- Laser Exposure Time:** 1.000 ms
- Grid spacing:** 15.0 mm
- Number of Planes:** 2
- Robot-Held Cal. Grid:** Yes
- Robot Holding Grid:** This Robot (Group 1)
- Cal. Grid Frame:** UT (1)
- Projection:** Perspective
- Override Focal Dist.:** No
- Min. Laser Contrast:** 50
- Min. Num Laser Points:** 50
- Max. Line Fit Error:** 1.5 pix

At the bottom of the window, there is a toolbar with buttons: **[TYPE]**, **LIVE**, **SNAP**, **FIND**, **END EDIT**, and **STOP**.

[illegible]

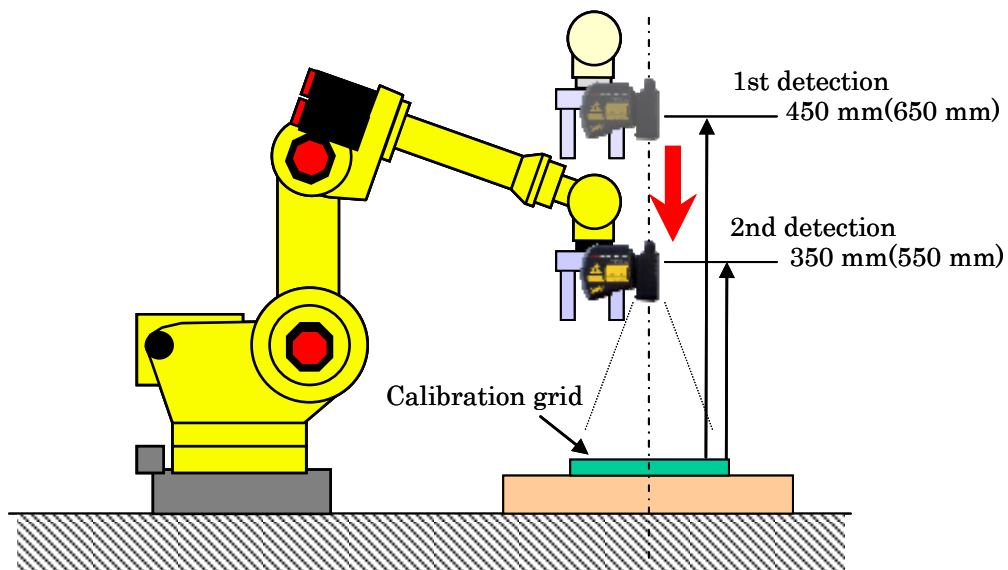
Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

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3DL CURVED SURAFACE SINGLE VISION PROCESS

7.2.4 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a robot-mounted camera, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



After one set of calibration tool is created with 3DL vision calibration, another set of calibration tool does not need to be created even after the camera Measurement Position is changed. This is because *iR*Vision uses the current robot position when calculating the position of the workpiece.

The teach screen for 3DL calibration is shown below. A calibration grid image is displayed.

[illegible]

Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

3DL Calibration Tool

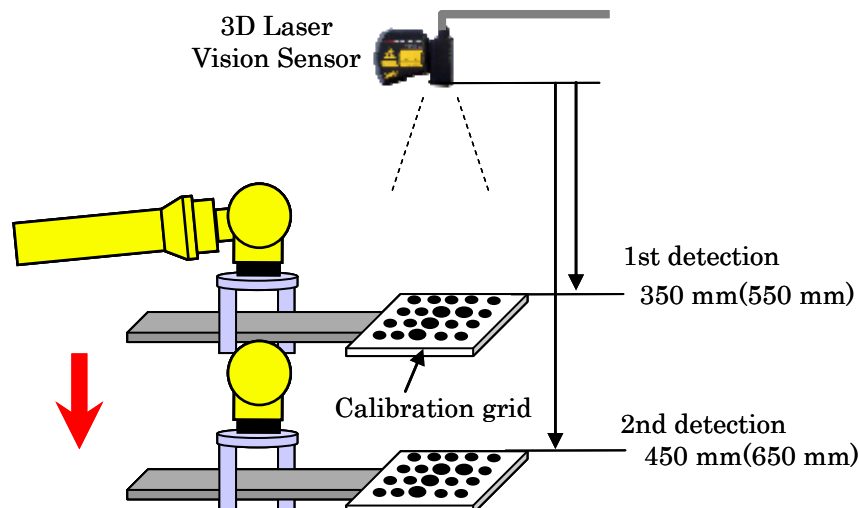
Setup	Data	Points
Robot to be offset:	This Robot	Group 1
Application Frame:	User Frame:	0
Camera:	Trained	
	CAM3DL	
Light Output Type:	DO	
Light Output Number:	1	
Light Output Polarity:	Signal=On -> Light On	
Light on Snap Delay:	10 ms	
LED Channel:	(1)	
LED Intensity:	1	
Image Display Mode:	2D Image	
2D Measurement Setups		
Exposure Time:	12,000 ms	↓ ↑
Light For Snap:	ON	
Laser Measurement Setup		
Laser Exposure Time:	1,000 ms	↓ ↑
Light For Snap:	OFF	
Grid Spacing:	15.0 mm	
Number of Planes:	2	
Robot-Held Cal. Grid:	No	
Robot Holding Grid:	This Robot	Group 1
Cal. Grid Frame:	UF 2	
Projection:	Perspective	
Override Focal Dist.:	No	
Min. Laser Contrast	50	
Min. Num Laser Points	50	
Max. Line Fit Error	1.5 pix	
Fixture Position Status:	Set	Set
1st Plane	Found	Find
2nd Plane	Found	Find

When recalibrating a camera, if you change the position of the calibration grid, set the user frame of the calibration grid frame again. Then click the [Set] button for the calibration grid frame and recalculate it.

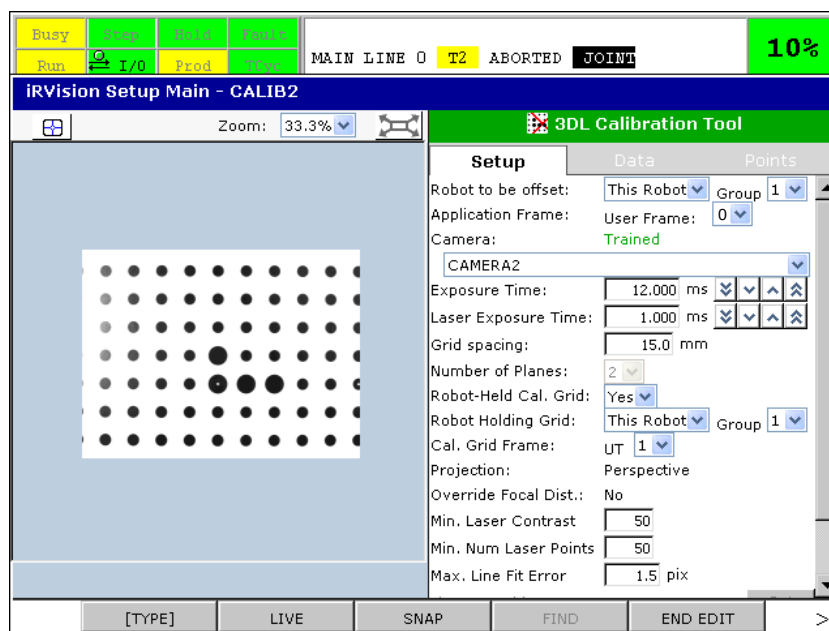
[illegible]

7.2.4 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a fixed sensor, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



The teach screen for the 3DL calibration is shown below. A calibration grid image is displayed.



						Title	Addition of the description about 3DL Calibration tool enhancement for FANUC Robot series R-30iB/R-30iB Mate CONTROLLER iRVision 3D Laser Vision Sensor Application OPERATOR'S MANUAL			
						Draw No.	B-83304EN-2/02			
Ed.	Date	Design	Description				FANUC CORPORATION		Page	34/40
	Date	2018.3.30	Design	T.Satou	Apprv.	T.Andou				

Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

3DL Calibration Tool

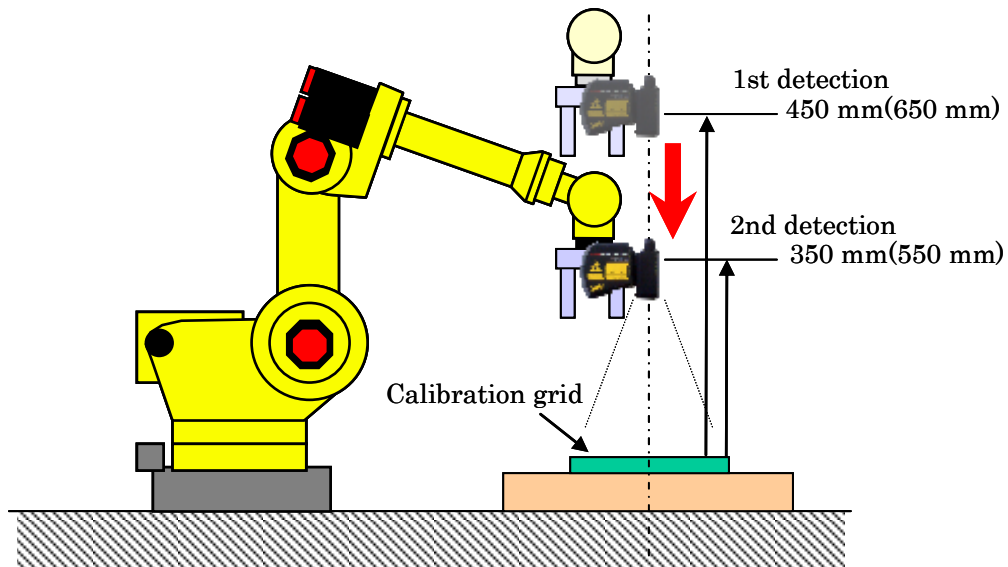
Setup	Data	Points
Robot to be offset:	This Robot	Group 1
Application Frame:	User Frame:	0
Camera:	Trained	
	CAM3DL	
Light Output Type:	DO	
Light Output Number:	1	
Light Output Polarity:	Signal=On -> Light On	
Light on Snap Delay:	10 ms	
LED Channel:	(1)	
LED Intensity:	1	
Image Display Mode:	2D Image	
2D Measurement Setups		
Exposure Time:	12,000 ms	↓ ↑
Light For Snap:	ON	
Laser Measurement Setup		
Laser Exposure Time:	1,000 ms	↓ ↑
Light For Snap:	OFF	
Grid Spacing:	15.0 mm	
Number of Planes:	2	
Robot-Held Cal. Grid:	Yes	
Robot Holding Grid:	This Robot	Group 1
Cal. Grid Frame:	UF 2	
Projection:	Perspective	
Override Focal Dist.:	No	
Min. Laser Contrast	50	
Min. Num Laser Points	50	
Max. Line Fit Error	1.5 pix	
Fixture Position Status:	Set	Set
1st Plane	Found	Find
2nd Plane	Found	Find

[illegible]

3DL CROSS SECTION VISION PROCESS

7.2.4 Creating and Teaching Camera Calibration Tool

To use the 3D Laser Vision Sensor, 3DL calibration using a calibration grid is required. For a robot-mounted camera, perform 2-plane calibration by moving the robot end of arm tooling up and down as shown in the figure below. For calibration, the appropriate distances between the 3D Laser Vision Sensor and calibration grid are near 350mm and 450mm (near 550mm and 650mm if the standoff of the 3D laser sensor is 600mm). When the calibration grid is detected with the robot and is made to have the same posture as it does when the workpiece is detected, the precision is increased.



After one set of calibration tool is created with 3DL vision calibration, another set of calibration tool does not need to be created even after the camera Measurement Position is changed. This is because *iRVision* considers the current robot position when calculating the position of the workpiece.

The teach screen for 3DL calibration is shown below. A calibration grid image is displayed.

[illegible]

[illegible]

Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

[illegible]

Change the distance between the camera and calibration grid, then click the [Find] button for [2nd Plane].

3DL Calibration Tool

Setup	Data	Points
Robot to be offset:	This Robot	Group 1
Application Frame:	User Frame:	0
Camera:	Trained	
	CAM3DL	
Light Output Type:	DO	
Light Output Number:	1	
Light Output Polarity:	Signal=On -> Light On	
Light on Snap Delay:	10 ms	
LED Channel:	(1)	
LED Intensity:	1	
Image Display Mode:	2D Image	
2D Measurement Setups		
Exposure Time:	12,000 ms	↓ ↑
Light For Snap:	ON	
Laser Measurement Setup		
Laser Exposure Time:	1,000 ms	↓ ↑
Light For Snap:	OFF	
Grid Spacing:	15.0 mm	
Number of Planes:	2	
Robot-Held Cal. Grid:	No	
Robot Holding Grid:	This Robot	Group 1
Cal. Grid Frame:	UF 2	
Projection:	Perspective	
Override Focal Dist.:	No	
Min. Laser Contrast	50	
Min. Num Laser Points	50	
Max. Line Fit Error	1.5 pix	
Fixture Position Status:	Set	Set
1st Plane	Found	Find
2nd Plane	Found	Find

When recalibrating a camera, if you change the position of the calibration grid, set the user frame of the calibration grid frame again. Then click the [Set] button for the calibration grid frame and recalculate it.

[illegible]